

Semiotics and Semantics (Lecture) ~ Lorena Sganzerla ~ Active Inference

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All right, hello and welcome everyone. This is the course Active Inference for the Social Sciences in 2023. It's August 30th or maybe a different date depending on where you are. And today we're in the Semiotics and Semantics section. We'll be hearing a lecture from Lorena Sganzerla, and then we're going to have a little bit of a discussion. So Lorena, thank you a lot for joining and off to you for the lecture. Okay, hello everyone. I'm based in Australia. Yeah, I'm Lorena. I'm doing actually my PhD in philosophy here. I work in cognitive sciences and philosophy. I work with him from concepts and notions of information. So I'm very interested in foundational understanding of cognitive sciences and information and how these kind of things cash out. And this is why I was interested in talking about semantics and social cognition to see how these things can show. I have a few ideas on how semiotics can be compatible with like active imprints. And I try to organize a little bit of this semiotic notion and active imprints to see if they pan out. Maybe they don't. So let's see how it goes. So let's talk about active inference, formal semantics, and social cognition. What's up there? Okay, I cannot move my slides. Now I can. So I will first have a very short introduction for you guys. I have out here. So what are you going to do? Like we have a short introduction. I'm going to talk a little bit of theories of meaning. And then we're going to talk a little bit the free energy of the logic principle and active inference. And how shared symbols and social cognition can be thought together. I bring one case that I find interesting and then we draw some conclusions. So let me know what you think in the comments and after we can have a little discussion or in the next session. So the background introduction that I want to first point to is we're talking about biological agents and how they act. So we're always thinking in terms of biological agents acting to express, in a way that express information of sensitivities to context dependent relevance to their environment and things that involve communities. However, how this very minimal way of talking about what motivates and what is relevant to people becomes stable enough as a self-sustaining pattern of interactions? Like are they stable and how they become that stable? So how can those information sensitivities ground values, norms, and goals consistently and over time? Assuming that those values and goals are necessary for meaningful interactions. So we have to align motivations to behavior, right? That's not really obvious. And ultimately how meaning is brought about? I don't think I'm going to answer that, but I want to point to some directions and how people have been thinking about these questions over time. So the problem of meaning has a brief account, right? Like traditionally approaches to cognition localize meaning inside individual organisms and more specifically inside their minds. So meaning pertains to I mean to content in this kind of thing. So meaning have, however, like meaning have a developmental history.

So it's hard to catch those two things together if it's inside people's minds, how we can have an evolutionary and developmental history. Assuming that meaning are things that we can learn over time and they happen in shared context. They don't happen out of nowhere in a vision from nowhere.

In a sense, like we can say that it's not really trivial that conventionalized words.

Like for example, meaning the conventionalized meaning of a word, for instance, cat means cat, have evolutionary histories.

The one thing does not come from the other. We need a broader understanding to cache this meaning out.

And that's where I'm stepping from. Like that's my grounding. Like we need to cache these things out in terms of community of language users.

Which brings a problem, right? We have goals and values that can be defined in terms of those communities and you have material and body factors.

So how these two things are together or how goals and values affect and motivate material and embodied agents.

To answer this kind of question, like over time, there are a few preferred ways. Like some, I would do a brief overview of some theories of meaning.

They usually in terms of cognitive sciences and any philosophy of mind. That's what I'm interested in.

How they can get it out. They usually try to think of those things in terms of syntax. Like concerns with form as an abstract organizing kind of rule.

Is a type of grammar and how the order of things and how the logical unfolding of those symbols, how they become meaningful.

And we, but the syntax itself is not really enough to cache out meaning and semantics.

So then in that sense, syntax evolved into a field that we call semantics in which meaning and mental content are organized throughout this instructional syntax.

Then we have prepositional knowledge and truth conditions in which allows these organizational story of meaningful words to make sense and they're going to point out into the, into the outside world.

And we have a third option.

And we have a third option. It's not so new, but it's not so famous, which is semantics or more importantly, I will focus on bio semantics.

And they use it concerns with the use of symbols, like how they bring about meaning in terms of pragmatics.

And that will be a little bit different and we'll catch it out in different terms in relation to its peers, like semantics and syntax.

Well, syntax is pretty clear in terms of being rules, right? So what is semantics talk about? So talk about mental content, about intentionality, about mental representations.

In the sense we have semantics that have specific key concepts that reference truth conditions and denotations.

So if you have mental content and if you put these meanings inside the organism, it has to refer to something else and how this cognitive penetration happens.

Like how can meaning come about inside the mind and how that refers to something outside in the world and how this observational viewpoint of you thinking about something in front of you can be actually true.

So we need, we need truth conditions that denote this specific relationship and that is represented like it's in

terms of mental content.

Contemporarily there's like various ways of catching this out. Like, uh, it can, that can be functional.

That can be more an internalized mental specific type of representations grounded in vehicles.

It's a very rich discussion. And I don't want to go into all that.

I just want to point it out that science and neuroscience use those terms to study and to build models that relates to mental representations,

in terms of neural representations, because it kind of makes sense for us.

Like brain activity, you can see them in terms of what happens by studying human beings and by studying animals and how these mental representations relate to something that happens in the world.

You can see that one behavior elicits specific activities and they have a correlational relationship.

And that is highly debated, but there we go.

We have many types of catching out neural representations and concerning to this, uh, relationship of stimuli and the interpretation of the stimuli and how they motivate action.

So that's, uh, that's already a very much weaker claim in relation to mental content and things being inside organisms, but it is still in the same line of reasoning.

And there is also bio-semiotics, which I want to focus a little bit more because it concerns the use and interpretation of signs and symbols, which then also emphasizes by the nature, the embodied nature and the situated nature of meaning.

So you don't have meaning without those practices and without the use of embodied agents.

So we have to reject the idea that meaning are abstract symbols, or they're going to be something that has to be in your mind and refer to something outside.

They're going to only exist in this type of relationship.

So, and they can take many forms, including chemical signals, like visual clues, behavior, words, like the word cat, you can have the, and how you organize the word cat.

They are all symbolic and that fits also in that kind of structure.

So they are very important for communication, obviously, because we do communicate in terms of pointing to the same shared notions of symbols in words and also other types of symbolic meaning.

And we coordinate life around them.

So living systems are then here characterized by the bio-semiotics, by their ability of creating and interpreting and using the symbols in terms of their practices.

The fathers, like the, of bio-semiotics, started like in, but probably in the 19th century from, and Ux Kohl is one of the most famous representatives.

Currently, you can find new authors like Hoffenmayer or Baker.

So I try to organize a little bit in a specific model, in a type of model, in a visual story, how these two types of understanding meaning are being applied in terms of theories of meaning historically.

So in bio-semiotics, we have the pragmatics and you have the practice and they relate to dynamics and context dependent.

So you can see here on the top of the screen, how agent, we have the agent relationship with its psychological and social aspects.

And bio-semiotics is somewhat concerned how correlation leads to prediction and beliefs and that leads to

expectations.

And we also have other types of theories like bio-semiotics and theories of computational syntax that tries to understand the dynamics and the mechanism of this relationship.

[illegible]

The free energy principle and active inference, as we know and have been talking about probably in the past two classes,

it's a model that is interesting for that because it has very minimal assumptions.

So it starts off from the idea that organisms act to maintain themselves in their expected biological and cognitive states.

So you're always talking, you're modeling and you're using these mathematical to ground biological agents in a very embodied sense.

And they do so by minimizing their free energy, right?

So they have these metabolic states that have to be minimized in terms of expenditure, in terms of uncertainty,

given that the long-term average of free energy is entropy, they want to minimize their uncertainty in terms of their metabolical states.

What that entails is that they have a specific range of states they want to stay in, like minimally to stay alive.

To stay alive, you have to do a bunch of things as a human being, but every single embodied agent would prefer to stay alive.

So there are some preferred basic states that will define how action will be taken first to maintain in metabolic states

and how that coordinate with the environment.

Which entails that organisms and agents can be described in those terms,

like an active inference tries to bring out those minimum assumptions in mathematical terms,

that try to express how reducing the probability of these non-anticipated states.

or states that are not favorable to maintaining their minimal metabolic needs,

will be reduced, right?

And that can be cashed out in terms of a generative model,
as we've been saying throughout the past actions,
how this generative model sets specific times and organizes around specific parameters
that can be described as trying to maintain those parameters within reasonable range.
So by minimizing free energy on average and over time,
the system will self-organize.

So we'll bring out the parameters of its internal states
that occupy a limited number of possible states.

We will set into a specific range that will then be,
that will be used as somewhat the base and the guide for this behavior.

You have two kinds of dynamics happening here, right?

That's what I want to talk about a little bit.

So we have then a phenotype that exists in a specific type of states,
but that will have to coordinate two different kinds of dynamics.

They're going to be stated here as complementary, right?

You have flow states and you have belief of data.

So with flow states, you have one type of modeling,
which is the non-equilibrium steady states.

Then we use intrinsic, it's named here intrinsic information geometry,
and you have belief updating that can be treated as extrinsic information geometry.

And those are different types of modeling
or the different types of thinking about those processes.

That's most important, right?

So in terms of flow states, we can talk about dynamical system
and how you settle and self-organize specific parameters.

And in terms of belief updating,
we will be talking about Bayesian mathematics
and Bayesian cognition
and specific ways of caching out.

How an agent will change and behave over time in relation to the dynamics that
is set into place in this state space of possible actions.

So while dynamical systems will explore temporal and contextual dependence between environment and agent,

we will set this specific state space that has temporal dimension and has contextual dimension.

And once this state space is sort of established,

we have more or less,

it's a more or less deterministic approach
to what can possibly unfold.

So how changes and how behavior

can be cached out in terms of non-deterministic behavior.

And that is defined by the different type of probabilistic thinking,
which is the Bayesian mechanics of belief updating.

So we have then a process of belief updating

throughout weighted interactions within this range of possible states.

So we do have dynamics and we do also have Bayesian process
unfolding together in terms of mechanisms.

And with that, we do have mathematical statements

providing a unifying perspective on formal approaches, connecting behavior and dynamics.

So that's what active inference should be bringing about

as a novelty in relation to older forms of understanding organisms,

which can either cache things out in terms of behavior or in terms of dynamics.

So trying to bring these two things together is quite interesting to understand
how symbols and what is semantics, for instance.

So that's the question I will be asking right now.

Let's talk a little bit about that.

I'm drawing here from a quite interesting paper from

Maxwell Halmstedt and Inet DiPolito.

Is free energy principle a principle for formal semantics?

So what would be formal semantics?

It sounds like an oxymoron, right?

Like how semantics can be formal.

How can you formalize it?

So semantics is the theory of meaningful things.

So how can that be empty of content?

Well, it has a few principles in which you can
explain what the formal semantics should be understood.

First off, we have Bayesian formalism that does not necessarily have to commit
to classical notions of representations, which are propositional in nature,
have propositional semantic content.

So the idea is that you can use the notion of a representation
in a way that is not necessarily contentful.

So how that's going to be.

So under the free energy principle, representations can be formal.

That's the argument they are bringing.

Under the physics of flow, specifically dynamical system theories and

inclination geometry, you can have these mathematical statements that only represent

possible specific ways of interacting how these correlations will affect each other in over time in this variational based type of cashing out mathematical functions.

In the sense, we can talk about representation, how we can be understood under this formalism.

And what they are saying is that they are best understood as internal structures that enable the system to parse out their sensory sensitivities or the sensory streams.

So to organize them in specific parameters and also in terms of their own Markov blankets and in their blanket states.

So that's which is interesting here, which is the new denovity of this paper, is that grounding the mathematical model in these mathematical properties of the system

will avoid grounding these properties in terms of the properties of symbols and thematics in abstract or subjective notions of meaning,

and the impossible ranges of action.

So summing up what we had before was a type of computational model that had specific components that does not allow for the intelligibility

of a cognitive component, which is intention, which is motivation, which is

to understand the context of brain-computer metaphors.

We have neuro and cognitive activity often seen as a transfer of symbolic information.

And cognition then is a snapshot of this transfer, so it's totally dependent of that kind of symbolical transfer, which is really difficult to explain how that's going to be biologically plausible.

So, and there's a lot of effort trying to do that, so if you're interested in this, there's a very vast wealth of literature,

but they have to answer to this problem.

So, how the mind and brain needs to produce an impromptu response to inputs of the external environment all the time.

And the idea of this formal semantics, the author is trying to explain,

is trying to include this component into the model.

So you have a deflationary model about content.

So you have a formal semantics, you have semantics without content, interestingly,

and formulated as a variation of free, a variation of free energy function.

So those structures that embody those processes correspond to states in the system.

So you have a very important thing.

And we also have the ecological component that can be formalized in terms of this two informational geometry that we, these two densities, the probabilistic densities, one from the dynamics and one from the Bayesian

describing behavior.

So how can active inference be compatible to biosimantics?

Can it be actually compatible or is it even necessary?

Well, active inference is a very general framework and then it does not really give us

the tools to understand how encoding can happen in terms of symbols, like how they can be broadcasted into

collective action.

Because prediction alone does not explain how do they come about.

So in this sense, we do have meaningful correlations that refer to the dynamics of one or two or more systems in terms of informational geometry that needs to be cached out with predictive value.

So how is this possible?

Well, we can talk about that in terms of shared attention.

That's one form of this kind of model to cache this out.

So we have active inference capturing relevant statistical relations between the environment and in aging.

And why we are, why I'm trying to think about here is how biosimiotics offer tools for this informational sensitivities and statistical relations to cache symbols in terms of use and interpretation within communities of language users.

And that can only happen in the sense via directed action.

So the problem now will turn into how is it possible to specify the factors that are sufficient to constrain action and direct that into specific
uh

and through the history of interaction.

Fout is European forestry.

So I'm picking this example from a very like the state from JNC's

I really recommend kind of
tells the story about how coordinated action,
how state coordinate specific collective actions.

But that's not the point that I want to talk here.

So here, let's unfold that a little bit in terms of the design of scientific forestry and think about scientific forestry as a practice.

And a practice that has an organizing principle that will like
um well that will be the main important uh articulating story that can connect together
how agriculture layouts plantations in specific ways and how collect collective farms will make specific choices in terms of this organization and how villages can structure themselves around
one specific notion or one specific organizational principle.

And how everything can be calculated to make one type of practice which is agriculture is very general more legible and more manipulable for different kinds of communities.

So in that we will involve manipulating the very concrete aspects of the land, the terrain, the products, the workforce.

So it has a very encompassing effect over time and in a very concrete
um in a very concrete sense.

So it's a very important thing that we have to do with the terrain.

And interestingly when you talk about European forestry and how

European forestry was modeling the terrain and how it's catching out value of these terrains,

they're not trying to do that to successfully depict the full range of diversity

of the actual social activity and the actual activity of agriculture or the richness of the vegetation.

Or they are not even trying, they didn't intend to do that.

That was not the point of this kind of modeling turning to a practice right.

So those models generally represent specific slices of nature that were interesting for one specific official observer.

And that point is really important because that will shape how all these communities and how the terrain and how the forest will unfold over time within this specific perspective.

So I bring here a quote that I really, really like because it tries to flesh this out in simple terms and very useful ones.

So thinking that exaggerating only slightly, one might say that the interest in forest

is not a specific single number, which was the revenue that yielded from the timber that was extracted annually.

Well, that's okay.

That's understandable.

Like that's intelligible.

And that happens still every day in terms of how we understand our economy.

But what is interesting about this quote is that the best way to appreciate

how heroic was this constriction of vision is to notice what falls outside the field of vision.

So what is missing from that is not the organizational principle of timber being valuable, but everything that was not there or was being excluded.

Trees and bushes and plants and plants that hold little to known potential for that purpose.

So missing out of that is flora diversity, folia, fruits, etc.

Twigs and branches.

Missing out of that are insects and small animals depending on the of the diversity.

All this ecological organization falls out of the organizational principle.

And that has very concrete consequences that we still suffer.

And we still live in the height of this climate crisis that also starts from this kind of thinking.

So it has really very material consequences for us.

And had for them at the time, if you read more about it, you know that the timber by like by trimming and domesticating nature so much like decreased in quality and then eventually decreased in value, which prompted them to try to reorganize that kind of story.

And what I want to call attention with the example is that also it brings some specific ways of talking

about it.

It brings vocabulary that only exists in terms of this type of practice specifically.

So this vocabulary constrains nature in a way by pruning what can, what cannot be used.

So it generates more than a legible nature, but a way of making nature intelligible and intelligible in a very specific certain way.

And we make that happen in terms of symbolizing things in specific terms.

But the symbols, they don't exist in themselves.

They exist only in organizing this social practice around them.

So when you replace nature with a term called natural resources, we are doing exactly that.

So it's specifying from a generalized notion of nature, like flora and fauna becomes something valuable, becomes a commodity.

But not everything, right?

It trims the practice.

It separates competing species from valuant.

So plants are symbolized as crops.

And what is, what the competing species are symbolized, the weeds.

They are not necessarily crops or weeds or anything like that in nature, but they are that for the way we talk like that because we practice agriculture in a certain way.

That also helps.

That's the same for insects or small animals that consume these interesting crops.

So the material effect of that and how that only gains traction in the practice over time and how we have specific intention that has unintended consequences are necessarily here situated, contextualized and embodied in terms of trimming specific types of attention.

And this attention will only shift to try to repopulate insect population, for instance, when you realize that this practice like brought this specific effect in a certain way that affects the value of our understanding of nature.

And that generates exactly a type of social coherence and social coordination and how constraints create this coherence, right?

So, let's talk back again about formal semantics.

While this formal semantics and inactive inference sort of embeds one type of mathematical formalism, symbols can only be grounded in specific concrete practices or in action.

So, while active inference can be used as a tool to explain social understanding, it does do that in terms of generalized synchronization,

In a model of cognition that can be inactive and dynamic, but is still very general.

In terms of bio-semantics, we can think of it as a framework that might be compatible with active inference in terms of explaining the materiality of the environment.

In a non-reductive and non-reductionist way that tries to retain some naturalistic credentials.

So, how that conventions are affecting and effecting the embodied agent in terms of forming those symbols

So, how that means?

but it will be a result of a specific participatory construction of the social scene over specific organizational principles and actions.

That allows us to say that social actors in this sense can behave as if they have a generative model that synchronizes with that symbolic state of affairs, allowing them to understand each other's behaviors or to develop approximates as informational sensitive or responsiveness to the same symbols and goals.

So in this sense, we don't have ever a view from nowhere.

That happens in the community, like for humans, at least it happens in community of language users.

So within a specific community of language users, some types of actions will be desirable and will be deemed as true in that specific type of community and can be only evaluated under those terms.

And you can talk about how rational that action is in terms of the conditions set by those communities.

So, in all scale of behavior, and that's the interesting thing.

So, when you have the problem of symbols, like in these two decoupled possible community of users and symbols and embodied agents and things that do not combine, we can try to model our agents in a way that we don't have this problem anymore, because you're trying to include that very, that very,

that is possible.

I don't know if that answers your question.

So, wondering about that fundamental question and then coming across active inference or, how elsewhere have people brought in the more formal and technical into this question?

What do you mean by more like formal and, can you repeat that please?

Yeah.

How does active inference help us or have continuity or difference with previous ways that people have connected the technical part from the, yeah, what you described as the formal semantics embeds the mathematical formalism and then the symbols are grounded in specific action.

So, has that been played out in other technical frameworks?

Well, we do have technical frameworks that try to organize that in terms of inactivism, right?

And autopoietic and activism.

But then we just have in these terms, the notion of symbols sometimes get, get loosely described.

And I think these guys are not really interested in that.

I think they're trying to describe something else, for example.

So, I think, if you're a little bit of a computational brain and computational mind metaphor, then you have a hard time also to explain how semantics are built into that computation, right?

I don't think anyone has a very strong answer to that.

You have to really buy into the idea of information processing.

And that's something that I'm, that, that's what brought me a little bit to that thing.

If you buy into the metaphor of information processing, then you need to have semantics within the brain.

If you try to lose a little bit this notion, if you try to understand process and information in different terms, what happens?

I think that was more or less the question that I was trying to ask.

And effective inference and biosemiotics can, because they have less assumptions in terms of what happens inside the processing information content in the mind.

If that can, if can, if that can bring interesting like answers, or if you can use as a model to organize behavior.

And, and there's another point that also brought me to this type of thinking was trying to understand how active, active inference can include his, the historicity of the agent.

Right.

Because I think that something comes across debated in active inference, how you bring novelty into the state space.

Like once you define that state space, the state space is more or less the feministic.

Because everything will happen within the specific range.

So how does novelty comes within this type of modeling?

So how can you explain that?

If, if it does, I don't know, it's more like, it's, it's more like an exploratory question that I was trying to make and see how that pans out.

I don't, I actually don't have an answer for that.

But if you take seriously, these act, some more inactive claims that you need history of interactions to, to understand how agents behave.

We cannot understand agents without that.

So how does active inference answer that these questions specifically?

Right.

Because once, yeah, that's just, I think that's a few kind of debated.

I'm not sure if there's new developments in this specific topic, but I remember, I think two years ago was like this specific papers from DiPaolo.

I think the forking laying down a forking path is specifically focusing on that and the formalisms and et cetera.

So I don't know if these two things can bring a little bit of more to the conversation.

Maybe not.

This is great.

I'll just ask some, some more general questions.

You're, you're adding a lot with these thoughts.

So you said that social interaction in this account is not reduced to prediction.

So how would you unpack that?

And how would you relate this?

I don't know exactly what term of art would go here, but this kind of social epistemology that active inference provides us in contrast to say predictive processing.

So how is it that, that we would or wouldn't reduce something to prediction?

And then what is it that active inference brings into that discussion?

Yeah.

Well, I think I have two points in there.

Like active inference in that sense has like our type of modeling that tries to include all the time and everywhere the embodied agent.

Right.

If, if you take seriously the idea of them and the model, the generative model is always accurately modeling that specific agent.

And because you can, it cannot be true or false that the model is your current state.

Like is like that.

Like, even if you are not feeling really well, you just know that you're not feeling well in relation to something else.

And that, that story is still very accurate to your state's condition right now.

Right.

So the prediction against traction in socially, at least in that sense, not in this specific detail of like predictive, predictive processing, I think, because I think that's more complicated.

But socially in social cognition, at least we can talk about this predictive value.

I think it's a more interesting way to do that because the model, the, the, the, the example that I use in terms of catching out symbols, we are using something that was meant in.

To make something intelligible that have very bad consequences that were not even close to be predicted.

So in that sense, and how these consequences happen.

Don't prediction cannot be there everywhere and all the time.

The prediction in this has an orientation as well.

Right.

So that's going to unfold only in terms of the orientation, but that's not all encompassing.

So, well, what does that say about the prediction?

That's why I was trying to think of it as not being a primitive, but something that comes along with this dynamics.

But yeah.

Well, this is a very deep, imprescent point that because the causal structure of the world changes, historiosity, agents change, they learn and they attend.

And then the niche changes with stigma and niche modification.

So because things really do change through time, then just training a data set at one time point.

And then using that same generative model and carrying it forward.

It's like picking a speed in a direction when you're on a winding road and just saying, all right, now let's turn on cruise control right now and just keep going in this direction.

It's working right now.

So then to connect that to the legibility of the forests as the causal model changed and externalities were not taken into account and so on, all these different features you brought in.

So now is active inference and cognitive modeling, cognitive technologies increasing our legibility of perception, cognition and action through surveillance technology and other mechanisms such that there could be slash we expect there to be also hidden dynamics and externalities.

And then we're also trying to see when we make some aspects of cognitive systems legible in the way that research in the neurosciences is happening towards.

Yeah, I think in a sense, like if you're trying to do that, you cannot have a knowledge on like a model that encompasses everything, right?

That would be non informative in terms of what you do when you do that, you have one interested thing.

You're interested in something, right?

You have one thing in mind when you're trying to flesh out in terms of cognitive modeling, how that, so how that applies to these more encompassing situations.

I think that's like that model is how I'm trying to ask this question.

So you can model them in this like sliced sense.

And that's, that's what we, that's what we do as scientists.

And that's how we learn about things.

But that only way of seeing that, that, that's very specific model is not all encompassing for all answers.

Like, so how can you, how can you use that in comparison to other situations?

So how can you integrate this, this type of thing?

I don't know.

I really, I don't have a question for an answer for that.

I don't know if we do have an answer for this, this type of question currently, but I think it's a very valid question to ask because, because also in the terms of sometimes applying that.

So, that sort of, that sort of, in a way, and that it becomes an organizing principle that becomes somewhat rarefied.

So keep an eye for this kind of metaphorical use, like assuming a little bit that all this talk can be metaphorical and it can be interesting and useful to catch them out, catch them out in these terms, because we don't have other ways to talk about them.

So, how we don't rate them and how they are, you know, because what I'm, I think I'm getting, I'm having a hard time to get there, but I'm thinking, I think I'm making the point of people criticizing the causal primacy.

So you, when you reduce this complex systems to one single cause, caching out everything that organizes the principle, this, this, this complex systems and one primal cause.

Then you are losing the complexity.

However, if you don't do that, how can you slice it out to understand the system?

So I think that's the two tensions here that I'm kind of trying to, to, to, to propose as a, as a question in turn, instead of an answer.

I don't know if that makes sense, but I think.

Yeah.

I hope people continue that discussion.

I'm going to build on that just to kind of maybe circle around that because it's really important.

So in the forestry case, the, the realist perspective, again, using retro casting the way we, we, or contemporary discussions, some of them in the active inference and free energy principle space are the realist says that region of land is just economic potential.

It is a natural resource that becomes a kind of handle.

The instrumentalist perspective takes a step back and says, it's as if that piece of land is a natural resource.

So yeah, the territory is indescribable and there's still a space for mystery, but then when the proxies come out because policies are being selected, the as if in some ways carries similar consequences to the actual, because they, and they may totally agree.

And, and the realists and the, and the instrumentalists might be like, yeah, it is economic potential or represents, or it could be treated as such, or it could be engaged with as such like that cognitive diversity can align on the social policy that then connects to these broader social questions and histories.

And so it's kind of interesting, like what work does this instrumentalism do?

Like, Oh, it's just a principle.

What work does it do?

Yeah.

And what doesn't it do?

And the ways that, that science and technical engineering fields are, and aren't deployed alone or together with other considerations.

Yeah.

That, that's very interesting.

Like in the sense that either if you're a realist or instrumentalist about that, like, if you think like models are not either true or false, they're just models, right?

And they might have views and it might, they might be capturing real properties, but the problem is in this conversation, even when you push them, like have realists and have an instrumentalists try to like loosen up a little bit.

And then you're just assumptions.

They still come, they still end up refining those principles and taking them to be more or less true or false.

Right.

Well, that's kind of the plot twist.

It's like you're appealing to what?

Surprise minimization as unifying the diversity responses that the realist and instrumentalists do.

So then does that make surprise or cognitive inertia like the ur principle where the maze just ends and it explains itself or how does that play in a pluralistic scientific setting?

Yeah, that's the, that's the, you, you, you, you might, you can have a third answer to that.

Like, right.

I have two.

Well, I think the answer that I tried to give is more eliminativist in the sense that eliminates the causal primacy of them, of these two modes of answering.

So when you think I'm thinking, I have in mind now, like actually just to try to make sense of that.

When you, when you think about biological information, you can think about biological information as being real world, as being metaphorical or instrumental.

Or you have these guys like Susan Oyama or Richard Lewontin, they try to bring some, they are against this cause of primacy.

They try to advocate for systems that do have that, but they have like a lot of more going on.

So that, that cannot be the only answer because those systems are not everywhere scientifically.

They don't have to be everywhere that.

So, for example, going back to the, to the forestry forestry.

So we really is going to say that territory is valued property.

And the, the, the, the, in the sense that instrumentalists concerned it look as if, and it can be used as if, but it's not necessarily value property.

It doesn't matter.

You can say like, well, this terrain is.

Seen from one perspective as value property, but it also has other uses around them.

So you had to be more inclusive of this diversity because they had this, the only organization principle around that specific terrain is not only to generate value.

Or if it does, this value doesn't have one single output.

It can have more than one output in terms of very specific terms.

It can catch it out very materially.

This terrain that has to provide monoculture needs to also, has to provide also value for the diversity of the fauna and the diversity of the flora around it.

And that is not included in a model because that's not what the model is done for.

The model has only specific use and that's fine.

However, that does not reduce the terrain to that.

And well, when you're talking about cognition, that might be more complicated.

But I think it's easy.

It's an easy way to talk about this causal primacy when you talk about forestry, because we are very much now experiencing this climate crisis.

And we see these things happening right now and how difficult because this type of thinking is built over time and so well fixed in the way we see things.

And we are understanding how hard it is to change that.

It's a paradigm shift that needs to happen for that to change.

It's not necessarily the model that is right or wrong.

It's just how you understand and apply them.

So there are pragmatics into that.

It's not only an instrumental, but it has something that the instrumentalist take does not capture.

And I don't know how to cast it out in one single word.

But that's what biosemantics was offering.

Biosemiotics was offering the idea of practice and pragmatics.

I don't know if it's the whole story, but it's another point into that story.

A lot of great points all connect that note of pragmatism to a little bit of the active inference formalism.

So when it comes to policy selection, behavior selection, some, but not all cognitive entities can act as if they're calculating expected free energy.

So these are the active, the truly active entities, whether they're simple or whether they're sophisticated and they do planning.

Expected free energy is how those agents are being modeled as if they're making decisions.

And it consists of the pragmatic and the epistemic value.

That's how these different policies are evaluated.

And so I see two angles or windows that that supports, at least from my non-social sciences background, a new kind of meeting in the middle there.

So first is the expansion of pragmatic value under the principle that surprise minimization or bounding is the imperative.

Right there in the pragmatic value, it doesn't have to be just body temperature or financial income.

We can expect and prefer ourselves to be learning new things or meeting new people.

And so we can load into the pragmatism different goals that under a narrower definition of utility or pragma have not traditionally been included.

So I think that first way is it speaks to a more inclusive concept of utility.

You might say, I'm also de-risking on forest fires alongside the timber and already you have now a plural

pragma.

So I think that's the kind of crossover.

And then the deeper and second and not saying there's only two ways that the conversation continues.

We have epistemic utility in terms of epistemic value.

And so the balance and the weighting between pragmatic value and epistemic or learning oriented activity overall helps control a dynamic balance with just like how tight is the grip and the preference overall.

And maybe we have some preference for timber to be exported from a region.

But we really don't know.

We don't have long term modeling.

We don't have deep causal analysis.

And so we're going to put 5% on the pragmatic extraction and other pragmatic features and leave the rest of the policy in an exploratory mode.

And come to a blend that's not just on the spectrum from explore to exploit, which is how this has been approached in like cognitive sciences previously.

It's like this is the third thing where policies that are on the frontier of exploring and exploiting rise to the top through their contributions to pragmatic and epistemic value.

And we made the model.

So that isn't to say that that is what's happening on the ground or maybe that's just the instrumentalist perspective.

Yeah.

Yeah.

Yeah.

Yeah.

But if it's an instrument, an instrumentless perspective might be a good one, right?

In the sense, the idea of inclusion is kind of the interesting one, right?

Because if you really find a notion of maximization, because you can use that in certain ways to understand what happens within, when you're modeling something, what is the limit of that maximization?

So that's, you can model agents as if maximizing utility, for instance, but maximizing utility is something that needs to be balanced out with utility of everyone around them.

Right.

And that's not trivial how that happens, right?

And even if it does, and even if it were trivial, it's not something that maps out completely identically with, oh, every single agent.

They don't, that's not, that there's not going to be, there's going to be diversity there.

So how do you make that notion more inclusive and less rarefied in that sense?

Because then if the more you rarefied, the more you single it out into around only one general type of understanding.

So when you think them more individually, then you kind of lose, you lose that strength of this one single principle.

Organized like the one single way of catching out utility.

That's not the principle, sorry.

So I think that's like speaks a lot with like the type of instrumental use and the type of pragmatism or how the time, the types of application of this practices, right?

Remember in the, in, in the sense of the scientific forestry, it's also a practice.

It's not something that is true in itself.

It's true in terms of practice, right?

In terms of applications, in terms of building communities around them.

So without this use, you, they, they have no grounding.

Yeah.

One really strong and salient point that I got from the lecture was about generalized synchrony.

And it's almost funny that we have to advocate or delineate generalized synchrony, because the alternative, the absence of generalized synchrony is just like, kind of like an ideal gas.

It's like there wouldn't be regularities in society.

We wouldn't be clustered in space or in time.

We wouldn't have information and communication and we wouldn't have what differentiates a society or a group or civilization or a colony or whatever from literally anything else.

So generalized synchrony is almost like all of the space between nothing and anything happening.

And then it supports all these different scenarios.

We could have this kind of mind in this kind of brain and this kind of environment and this kind of niche.

And so we can talk about the ant colony foraging algorithms that work in the desert.

And there might be mutual, non-exhausted, open-ended change of algorithms and what works in a different setting.

And with humans, we have the sophistication and the history and the scope and the scale and all of these metas and weaving with is and ought and our own forward going direction.

And then we have the sophistication of the direction, which is really the moment where it goes from being a forensic sociology to the real-time sensemaking and action selection based upon our cognitive models in the broader sense, which includes humans and non-human agents.

But it's like, that's when it goes from being more than just an understanding.

When that understanding translates to an action policy and they do, then it matters as much as anything does.

Yeah.

One of the criticism of the active inference, I think, is like, it's too general to single out specific constraints that orient action.

Right.

That's what I saw, like, thinking about how symbols are brought about in community of language users.

Assuming that the symbols are not, they are only meaningful throughout these histories and practices can help us to apply this type of general story.

Right.

Yeah.

Navigating specific, well, epistemic communities.

And I think like that's the work that Mahou is going to bring along in terms of epistemic communities.

And then you single out in them.

So you have these communities of language users in which specific practices create specific orientations.

Yeah.

One thought on that.

I'm not asking like, what does active inference or linear regression or some other method.

Again, this is, who knows, is this just deep esoteric instrumentalism.

Yeah.

But asking, what does it have to say about this symbol?

What does active inference have to say about the letter E in English?

Can it be as simple as to say that that's a category error and that you could make a toy example or a pedagogical example where symbol one and symbol two oscillate in a multi-agent simulation?

Or you could make a generative model for an alphabet and leave it abstract, but you're never going to connect to the historiosity and the real enactment of the English letter E unless you're confronting the minute particulars and the actual embodiment of that letter.

And then there's different course grainings, different iterated modeling decisions, different questions about how do you include and participate in a different stage of the scientific process.

But it's like, you're in the specifics at that point.

Active inference doesn't have an opinion on the letter E because we wouldn't want it to.

It would be too opinionated about the wrong things.

It'd be making assumptions and bringing axioms into play that would be extremely constraining.

And so it remains general as a framework and the free energy principle even more slash differently so.

But when we get into the actual, that's where we're connecting the real systems of interest.

But how can we make claims and advance them socially about systems of interest when we're not even, it's like comparing bridge building as a concept with this bridge right here.

It's a little bit like a, a category disconnect.

Yeah.

Well, it depends also how you single out some things.

They're not, they're just, you just don't do that because they're just not interesting enough, right?

There's nothing like that comes out of it in terms of this bridge right here in a single.

How did you put it again?

Like your category of bridges in this bridge right here, right?

When you do that, you can learn something about the category in terms of this single one right here.

But it does not, it's not exhaustive.

That's the, that's, I think one point it's not exhaustive and it's not right or wrong.

It might be better than other, I think the idea like the, you can bring some, you can make some comparisons in terms of like predictive value.

You can have like better or worse ways of doing it.

And then the gate gains more utility.

Yeah.

Okay.

Connecting this to also this, this non-exhaustive nature and where the power of a generalizable framework is.

Let's talk about the category of golden gate bridges in San Francisco.

So there's only one actual bridge there, but we're generalizing it.

And so somebody makes a state space and says, well, I'm exploring if there was more than two end to end.

So they're exploring the state space of topologies of bridge.

And someone else is exploring the state space of color.

And somebody else is exploring the state space of traffic dynamics.

And someone else is combining two of them or combining all three of them.

And then that's the conversation is which map of this category of state spaces, category of adjacent possibilities, as we chose to model it.

And if you don't include traffic flow, don't be surprised that it's not going to take that kind of data or make that kind of prediction.

And some variables matter and some matter in combination and some don't.

But when we model, we kind of build these regions or adjacencies to the actual, which is where the historicity of the past leaves us.

And then we have an adjacent possible as modelers and sense makers, including absurd counterfactuals and state spaces.

And then the state spaces are mobilized.

Yeah, in a way. But if you talk about like specific type of engineering, yes, in that sense, they are mobilized and they are not true or false.

You can use false models that is to function.

That's not like you can have specific uses of things that you know, they're not really true.

Like the thing that tolerate, tolerate, tolerate type of modeling, the social, the solar system is to use that to navigate, you know, in the desert, even though it's not really accurate.

But if you want to go from A to B helps you to do that.

That's a different thing, right?

Like it is not true or false.

But when you think about agents in terms of like, and if bringing back to the idea of symbols being only cashed out in terms of community of users that have histories and they are made useful, that are made meaningful in terms of their practices.

Then you don't have ever disconnect, right?

You have a, you can think about this cognition is always connected to this specific types of practicing.

You can even single out some moments of it and try to see what you learn from this kind of slice of time.

But when you think about the whole story of the cognitive system, you can, if you're going to really think of it in terms that it's valuable, you cannot disconnect from the state, let's say the state space of practice.

Awesome.

Well, do you have any closing thoughts or what are you looking forward to with the discussion section, the rest of the course, just by way of wrapping up and letting people know any other thoughts?

Yeah, it would be cool to explore this, this idea that we've been actually bringing along here, right?

How we can go with that like instrumentally realistically or how can we understand this general synchronization of the model in terms of specific instantiations and how informative they are?

Is that really interesting or how can these things be actually reified in terms of modeling?

And if that happens, does that have any specific consequences actually or not?

Right?

Maybe not.

Awesome.

Well, very ambitious and excellent lecture.

So thank you very much, Lorena.

And talk to you.

Just keep in mind, I was just like trying to ask these questions.

I don't also have those answers.

Like it's, I found very little about active inference and semiotics.

Like produced, I found like a few papers.

So that's very exploratory actually.

So yeah.

So you're invited to think of these things and explore a little bit them further if it's possible.

Yeah.

Cool.

Epic.

All right.

Thank you, Lorena.

I'll see you in a couple of weeks.

Farewell.

See you.

Bye.

Bye.

Bye.

Bye.